

TWINKIT · REQUIREMENTS

TwinKit requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/twinkit		
Source	docs/03-requirements.md @ 4f7e44a		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R1 restated with adaptive data rate, R12 restated at light load, R14 target raised to \$300
0.3	2026-09-25	Amish Chadha	Status from the TRL 3 calculations (TWK-CAL-001); R1 found not met at SF9; decisions per TWK-DDR-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

TwinKit requirements

Status is judged from the TRL 3 calculation note TWK-CAL-001 v0.2 and the parametric model. No requirement is now shown as not met or at risk; R13 (setup time) can only be shown by a timed trial. Three requirements were changed at this revision by Amish's decisions of 2026-09-25 (TWK-DDR-002): R1 now requires adaptive data rate (was: under 1 % at SF7 to SF9, which failed at 1.02 % with every node on SF9); R12 now applies at the normal light load, with heavy jobs scheduled for cool hours (was: at any load, at risk); and R14's parts limit rises from \$200 to \$300 (was not met at \$290).

Table 1. Requirements.

ID	Requirement	Target	Status at TRL 3 (TWK-CAL-001)	Verification (TRL 4 or later)
R1	Receive readings from lab sensor nodes	50 nodes at 1 reading per 5 min each, under 1 % of uplinks lost, with adaptive data rate enabled on the network server so each node uses the fastest spreading factor (SF7 to SF9) its link allows	Met: 0.21 % for an SF7 to SF9 mix [A3, A6]; for information, 1.02 % if all nodes were forced to SF9 [A2]	Bench and field packet counts
R2	Accept common inputs	LoRaWAN (EU868 or US915), MQTT over Ethernet or Wi-Fi, HTTP POST	Met by design	Configuration check
R3	Store history on the gateway	1 year for 50 nodes x 10 fields at 5 min intervals	Met: 4.73 GB per year uncompressed on 56 GB free [B1, B2]; card wear 5.5 to 8.2 card cycles per year, to watch [B4]	Measured database size
R4	Link data to the model	Every sensor channel mapped to a named part (BOM number) of the project's build123d model and to an expected value or calculation, in a validated schema	Met by design (schema fields adopted, TWK-DDR-001)	Schema review on the FieldNode example twin
R5	Compare measured and expected	Residual and pass or flag state shown within 60 s of a reading arriving; flag when outside a stated tolerance for a stated time	Met: 0.9 to 4.7 s [C1]	Bench run
R6	Show the twin in 3D	The project's glTF model, colored by live values, in a current browser on a mid-range laptop or phone	Met by design; frame rate not verifiable at TRL 3	Browser trial
R7	Work offline	All functions run with no internet; optional upstream sync buffers at least 7 days	Met by design; 7-day buffer 91 MB [B3]	Offline run
R8	Low-voltage power	9 to 30 V DC input, 12 V nominal; average draw 8 W or less	Met: 6.41 W average; 20.6 W peak, 2.29 A at 9 V, 3.15 A T fuse [D1 to D3]	Power measurement
R9	Ride through outages	At least 30 min on backup and a clean shutdown before the pack is empty	Met: 2.40 h new, 1.63 h aged pack at 0 °C [E1]	Timed outage
R10	Compact mounting	Fits a TS35 DIN rail; gateway, converter and backup within 20 modules (350 mm) of rail	Met: 294 mm, 16.8 modules [H1]	Fit check
R11	Secure by default	Unique credentials set at first boot, TLS on the dashboard, no ports open beyond the local network by default	Met by design	Configuration audit
R12	Operate in a lab or equipment room	Ambient 0 to 40 °C without CPU throttling at the normal light load; heavy jobs (database recompression, backups, rebuilds) scheduled for cool hours	Met: processor 69.1 °C at light load, vented, 15.9 K below an assumed 85 °C throttle point [F6]; 96.9 °C if a heavy job ran at 40 °C [F2]	Thermal test at 40 °C
R13	Quick setup		Not verifiable at TRL 3	Timed trial

ID	Requirement	Target	Status at TRL 3 (TWK-CAL-001)	Verification (TRL 4 or later)
		From a flashed card to the first live reading on the dashboard in 60 min or less, following the guide		
R14	Low cost and buildable	Parts \$300 or less (raised from \$200, TWK-DDR-002); no custom PCB	Met: \$290.00 [I1, I2]; no custom PCB	Supplier quotes
R15	Open and exportable	All software open source; data exportable as CSV and through an open API	Met by design	Design review
R16	Monitoring only	No control outputs; the gateway cannot actuate the monitored system	Met by design	Design review

■ Assumptions

The assumptions behind each status are listed in TWK-CAL-001, Table 1. The main ones are:

- A sensor reading is a LoRaWAN uplink of 20 bytes of payload plus 13 bytes of overhead, at 125 kHz and coding rate 4/5 (71.9 ms at SF7, 246.8 ms at SF9, 1810.4 ms at SF12).
- Uplinks spread evenly over 8 channels and arrive at random (pure ALOHA), with no capture effect.
- A stored value takes 90 bytes including index overhead, before compression.
- The gateway draws 6.41 W on average: board 4 W, fan 0.5 W, concentrator 1 W, converter at 90 % and UPS loss 0.3 W.
- The backup pack delivers 80 % of its 19.2 Wh rating when new.
- The processor throttles at 85 °C and sits 4 K/W above the enclosure air with its fan running; both are assumptions to be measured.